

DLC Container Neuron DDOS

1. git clone and build AWS deep-learning-container

<https://github.com/aws-neuron/deep-learning-containers/tree/main>

Make sure to follow prerequisites and
Build container using following

`docker build --no-cache -f docker/pytorch/inference/1.13.1/Dockerfile.neuron -t matrix-mul-docker-image .`

2. From steps in

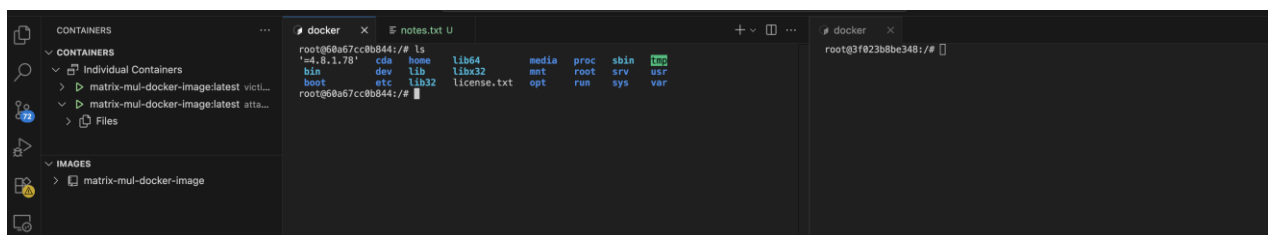
<https://awsdocs-neuron.readthedocs-hosted.com/en/latest/containers/tutorials/build-run-neuron-container.html>

Start two containers victim and attacker from below commands

`docker run -it --name victim --device=/dev/neuron0 matrix-mul-docker-image:latest /bin/bash`

`docker run -it --name attacker --device=/dev/neuron0 matrix-mul-docker-image:latest /bin/bash`

3. Once containers are running transfer cda.zip to both containers and extract them and install gdb
`apt install gdb`



DLC Container Neuron DDOS

Attacker container inference successful.

```
docker  X  notes.txt U
root@60a67cc0b844:/cda/matrix_mul# ls
aws_strace_codes.log  exploit.gdb  inferencecorruption.log  originaltrace.log  run_neff  strace.log  trace.log
compile.py            exploit_ddos.gdb  launch2inferences.sh    original_model.pt  run_neff.c  test.c      trace_all_neurons_matrix.log
compiled_model.pt     gdb.txt       ndl_memory_get_pa.log  pseduocode.c      saved_temps  teststrace.log  trace_matrix.log
ddostrace.log        inference.py    neuron_config.sh       pwntool.py        stopAtPa.gdb  teststrace_1.log  trace_matrix_core0.log
root@60a67cc0b844:/cda/matrix_mul# python3 ./inference.py
Attempting to load model from: compiled_model.pt
Model loaded from compiled_model.pt
Inference input:
tensor([[[[1., 1., 1.],
          [1., 1., 1.],
          [1., 1., 1.]]]]])

=== Running Inference ===
Inference output:
tensor([[[[8.4689]]]])
Virtual address of output: 0x7f396c8a25c0
root@60a67cc0b844:/cda/matrix_mul#
```

Victim container inference successful.

```
docker  X
root@3f023b8be348:/# cd cda/
root@3f023b8be348:/cda# ls
matrix_mul  original
root@3f023b8be348:/cda# cd matrix_mul/
root@3f023b8be348:/cda/matrix_mul# ls
aws_strace_codes.log  inference.py  pseduocode.c  test.c
compile.py            inferencecorruption.log  pwntool.py    teststrace.log
compiled_model.pt     launch2inferences.sh    run_neff      teststrace_1.log
ddostrace.log        ndl_memory_get_pa.log  run_neff.c    trace.log
exploit.gdb          neuron_config.sh       saved_temps   trace_all_neurons_matrix.log
exploit_ddos.gdb     originaltrace.log      stopAtPa.gdb  trace_matrix.log
gdb.txt              original_model.pt      strace.log    trace_matrix_core0.log
root@3f023b8be348:/cda/matrix_mul# python3 inference.py
Attempting to load model from: compiled_model.pt
Model loaded from compiled_model.pt
Inference input:
tensor([[[[1., 1., 1.],
          [1., 1., 1.],
          [1., 1., 1.]]]]])

=== Running Inference ===
Inference output:
tensor([[[[8.4689]]]])
Virtual address of output: 0x7f1b723525c0
root@3f023b8be348:/cda/matrix_mul# python3 inference.py
Attempting to load model from: compiled_model.pt
Model loaded from compiled_model.pt
Inference input:
tensor([[[[1., 1., 1.],
          [1., 1., 1.],
          [1., 1., 1.]]]]])

=== Running Inference ===
Inference output:
tensor([[[[8.4689]]]])
Virtual address of output: 0x7fd40e4d8720
root@3f023b8be348:/cda/matrix_mul#
```

DLC Container Neuron DDOS

Attacker container runs inference inside gdb and sources gdb_exploit script.

Note run inference inside gdb so it can get symbols and source the exploit script

```
root@60a67cc0b844:/cda/matrix_mul# gdb python3
GNU gdb (Ubuntu 9.2-0ubuntu1~20.04.2) 9.2
Copyright (C) 2020 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from python3...
(gdb) r inference.py
Starting program: /opt/conda/bin/python3 inference.py
warning: Error disabling address space randomization: Operation not permitted
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
[New Thread 0x7f0c557ad700 (LWP 50)]
[New Thread 0x7f0c52fac700 (LWP 51)]
[New Thread 0x7f0c527ab700 (LWP 52)]
Attempting to load model from: compiled_model.pt
Model loaded from compiled_model.pt
Inference input:
tensor([[[[1., 1., 1.],
          [1., 1., 1.],
          [1., 1., 1.]]]])

=== Running Inference ===
Inference output:
tensor([[[[8.4689]]]])
Virtual address of output: 0x7f0c4ad26570
[New Thread 0x7f0c4a1e8700 (LWP 53)]
[Thread 0x7f0c4a1e8700 (LWP 53) exited]
[Thread 0x7f0c527ab700 (LWP 52) exited]
[Thread 0x7f0c52fac700 (LWP 51) exited]
[Thread 0x7f0c557ad700 (LWP 50) exited]
[Inferior 1 (process 46) exited normally]
(gdb) █
```

Attacker source exploit_ddos.gdb script

[illegible]

DLC Container Neuron DDOS

Inferences will start to fail for all Victim containers even after attacker container is removed.

```
Model loaded from compiled_model.pt
Inference input:
tensor([[[[1., 1., 1.],
          [1., 1., 1.],
          [1., 1., 1.]]]]])

==== Running Inference ====
2025-Sep-24 22:43:15.0062 25:25 ERROR TDRV:exec_consume_tpb_status_notifications Missing infer_status notification: (endi:0)
2025-Sep-24 22:43:15.0062 25:25 ERROR TDRV:exec_consume_tpb_status_notifications Missing infer_status notification: (endi:1)
2025-Sep-24 22:43:15.0062 25:25 ERROR TDRV:exec_consume_tpb_status_notifications Missing infer_status notification: (endi:2)
2025-Sep-24 22:43:15.0062 25:25 ERROR TDRV:consume_model_start_extra_notifications_v1(FATAL-RT-UNDEFINED-STATE) model start timeout (2000 ms) on Neuron Device 0 NC 0, waiting for execution completion notification
2025-Sep-24 22:43:15.0062 25:25 ERROR NMG:dlr_well_start_no_lock Model 1002 start failed for VNC=0, ret: 5
2025-Sep-24 22:43:15.0062 25:25 ERROR NMG:tltps_infer_lock Failed to start model
2025-Sep-24 22:43:15.0062 25:25 ERROR NMG:dlr_infer Failed to acquire infer locks

Inference failed: The following operation failed in the TorchScript interpreter.
Traceback of TorchScript, serialized code (most recent call last):
  File "code/_torch/_torch_neuron/runtime/_torch_mangle_5.py", line 11, in forward
    _NeuronGraph_0 = getattr(self, "_NeuronGraph_0")
    model = _NeuronGraph_0.model
    _0 = ops.neuron.forward_v2_1([argument_1], model)
    ~~~~~
    return _0
    ~~~~~
HERE

Traceback of TorchScript, original code (most recent call last):
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch/_ops.py(442): __call__
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch_neuron/decorators.py(416): forward
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch/nn/modules/module.py(1176): _slow_forward
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch/nn/modules/module.py(1192): __call__impl
Open file in editor (cmd + click) 1.13_inf1/lib/python3.10/site-packages/torch_neuron/graph.py(580): __call__
1.13_inf1/lib/python3.10/site-packages/torch_neuron/graph.py(195): __call__
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch_neuron/runtime.py(69): forward
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch_neuron/graph.py(187): run_op
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch/nn/modules/module.py(1192): __call__impl
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch/jit/_trace.py(957): trace_module
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch_neuron/tensorboard.py(180): tb_parse
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch_neuron/tensorboard.py(154): tb_graph
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch_neuron/decorators.py(523): maybe_generate_tb_graph_def
/opt/aws_neuron_venv_pytorch_1.13_inf1/lib/python3.10/site-packages/torch_neuron/convert.py(164): trace
/home/ubuntu/cda/matrix_mul/compile.py(36): compile_model
/home/ubuntu/cda/matrix_mul/compile.py(14): <module>
RuntimeError: Failed to execute the model status=5 message=Timeout Exceeded

2025-Sep-24 22:43:15.0066 25:25 ERROR TDRV:model_stop Model stop request for model 1002 doesn't match currently running model 0
2025-Sep-24 22:43:15.0066 25:25 ERROR TDRV:kbl_model_stop Failed to stop model, errs: 3 for NC:0
2025-Sep-24 22:43:15.0066 25:25 ERROR NMG:dlr_well_stop Model 1002 stop failed for VNC=0, ret: 3
terminate called after throwing an instance of 'c10::Error'
what(): Could not unload the model status=3 message=Invalid Handle
Exception raised from 'NeuronModel at /opt/workspace/KamayaPyTorchRuntime/neuron_op/model.cpp:198 (most recent call first):
frame #0: c10::Error::Error(c10::SourceLocation, std::string) + 0x57 (0x7f0106fde457 in /opt/conda/lib/python3.10/site-packages/torch/lib/libc10.so)
frame #1: c10::detail::torchCheckFail(char const*, char const*, unsigned int, std::string const&) + 0x64 (0x7f0106fa33ec in /opt/conda/lib/python3.10/site-packages/torch/lib/libc10.so)
frame #2: neurons::NeuronModel::NeuronModel() + 0x51b (0x7f00f9f72ae0 in /opt/conda/lib/python3.10/site-packages/torch_neuron/lib/libtorchneuron.so)
frame #3: std::lisp::base64_encode(gnu_cxx11::Lock_policy)2>::M_release() + 0x46 (0x7f00f9fde1c5 in /opt/conda/lib/python3.10/site-packages/torch_neuron/lib/libtorchneuron.so)
frame #4: neuron::Model::Model() + 0x71 (0x7f00f9f69811 in /opt/conda/lib/python3.10/site-packages/torch_neuron/lib/libtorchneuron.so)
frame #5: neurons::Model::Model() + 0x9 (0x7f00f9f69809 in /opt/conda/lib/python3.10/site-packages/torch_neuron/lib/libtorchneuron.so)
frame #6: <unknown function> + 0x65d07b (0x7f015f1c507b in /opt/conda/lib/python3.10/site-packages/torch/lib/libtorch_python.so)
frame #7: <unknown function> + 0x65d07b (0x7f015f1c507b in /opt/conda/lib/python3.10/site-packages/torch/lib/libtorch_python.so)
frame #8: <unknown function> + 0x66e14c (0x7f015f1c014c in /opt/conda/lib/python3.10/site-packages/torch/lib/libtorch_python.so)
frame #9: <unknown function> + 0x99efb5 (0x7f015f408fb5 in /opt/conda/lib/python3.10/site-packages/torch/lib/libtorch_python.so)
frame #10: <unknown function> + 0x3c7f73 (0x7f015eeaf7f3 in /opt/conda/lib/python3.10/site-packages/torch/lib/libtorch_python.so)
frame #11: <unknown function> + 0x3c86cf (0x7f015eeb06cf in /opt/conda/lib/python3.10/site-packages/torch/lib/libtorch_python.so)
frame #12: <unknown function> + 0x3c82fb (0x7f015eeb02fb in /opt/conda/lib/python3.10/site-packages/torch/lib/libtorch_python.so)
frame #13: <unknown function> + 0x3c86cf (0x7f015eeb06cf in /opt/conda/lib/python3.10/site-packages/torch/lib/libtorch_python.so)
omitting python frames>
frame #25: _libc_start_main + 0xf3 (0x7f017df10883 in /lib/x86_64-linux-gnu/libc.so.6)
Aborted (core dumped)
```